MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATIONS, MARCH 2025
Fourth Semester
DSC-A – MG4DSCZGY200
ANIMAL DIVERSITY - CHORDATA I

Duration: 1.5 hr Maximum Mark:50

PART A – MCQ (12 × 1 mark) Write all 12. (12 X 1 =12 marks)

1. Notochord in chordates is derived from: [U][CO 1]
   a. Mesoderm
   b. Endoderm
   c. Ectoderm
   d. Neural crest
2. Homocercal tail is seen in: [U] [CO 1]
   a. Shark
   b. Lungfish
   c. Rohu
   d. Amphioxus
3. Cutaneous respiration means respiration through [U] [CO 2]
   a. Lungs
   b. Skin
   c. Mouth
   d. Gills
4. Reptiles without temporal fossae in the skull belong to: [U] [CO 1]
   a. Diapsida
   b. Parapsida
   c. Anapsida
   d. Synapsida
5. "Dry bite" means: [U] [CO 3]
   a. Bite without pain
   b. Bite without venom injection
   c. Bite causing bleeding only
   d. Bitten by a non-venomous snake
6. Number of vertebrae in frog: [U] [CO 2]
   a. 7
   b. 9
   c. 10
   d. 12
7. Adult frog is: [U] [CO 2]
   a. Ureotelic
   b. Uricotelic
   c. Ammonotelic
   d. None
8. Ichthyosaurus belongs to which subclass? [U] [CO 1]
   a. Anapsida
   b. Parapsida

c. Diapsida
d. Synapsida

9. Dipnoi are also known as: [U] [CO 1]
   a. Lung fishes
   b. Jawless fishes
   c. Cartilaginous fishes
   d. Ray- finned fishes

10. Male seahorse protects eggs in: [U] [CO 1]
   a. Nest
   b. Mouth
   c. Brood pouch
   d. External substrate

11. Which of the following is a venomous snake? [U] [CO 3]
   a. Calotes
   b. Dendroaspis
   c. Naja
   d. Python

12. Point out the feature for identifying a venomous snake? [U] [CO 3]
   a. Vertical slit pupil
   b. Rounded pupil
   c. Smooth scales on back
   d. Short tail

PART B- Short Answer (3 marks) Write any 6. **(6 X3 =18 marks)**

13. Briefly explain retrogressive metamorphosis. [U] [CO 1]
14. How do snakes bite and inject venom? [A] [CO 3]
15. What is the structure and function of the frog heart? [U] [CO 2]
16. Comment on Paedogenesis. Write an example. [U] [CO 1]
17. Write a short note on *Nasikabatrachus sahyadrensis*. [U] [CO 1]
18. Briefly explain the digestive system of a frog. [U] [CO 2]
19. Outline the classification of Class Reptilia up to Orders. [U] [CO 1]
20. Comment on Accessory Respiratory Organs in Fishes. [U] [CO 3]

PART C – Short Essay (5 marks) Write any 4. **(4 X5 =20 marks)**

21. Explain the types of parental care and their biological significance in fishes. [U] [CO 1]
22. Give a detailed account of blood circulation in frogs. [U] [CO 2]
23. Define migration in fishes and describe the different types with examples. [U] [CO 1]

24. Analyse the morphological characters used to distinguish venomous and non-venomous snakes. [A] [CO:3]
25. Write an essay on the non-venomous snakes commonly found in Kerala. [U] [CO: 3]
26. Explain the respiratory system and mechanism of respiration in frogs. [U] [CO: 2]